

Exercise 5.3

Copy the following into **main.py** and then read the rest of the Exercise.

```
def num_digits(x):  
    # remove the following line and write the function here  
    pass  
  
def main():  
    # remove the following line and write your program here  
    pass  
  
# This is needed for the tests. This is now the first bit of code Python will run.  
if __name__ == "__main__":  
    main() # calls the main function
```

Write a program that asks the user for a positive whole number. Your program should output the number of digits in the whole number. If the input is invalid the program ends with **Invalid input**.

Your program must be written inside the **main()** function and work the same as the given examples:

```
Please enter a positive whole number:  
3  
3 has 1 digit.
```

```
Please enter a positive whole number:  
345  
345 has 3 digits.
```

```
Please enter a positive whole number:  
345.6  
Invalid input.
```

```
Please enter a positive whole number:  
0  
Invalid input.
```

```
Please enter a positive whole number:  
Sam
```

Invalid input.

Your program must use a function `num_digits` that takes a `int` as a single parameter and returns the number of digits in a positive integer, `n` as a type `int`.

Hint. To determine the number of digits in a positive integer `n`, divide it by `10` repeatedly. When `n < 1`, the number of divisions to date indicates how many digits the number originally had.

```
345/10 = 34.5
Is this less than 1? No.
34.5/10 = 3.45
Is this less than 1? No.
3.45/10 = 0.345
Is this less than 1? Yes.

3 digits...
```

For example,

`num_digits(5467)` should return `4`.

`num_digits(100000)` should return `6`.

The skeleton of the program is set up for you. You will see the `pass` statement, this is just a placeholder for the function body. You should remove this and replace it with the correct code.